ELECTRODYNAMICS II · SEMESTER 5

Transients and alternating currents

Module 4 · PYQ + model paper

40 questions · 20 short + 16 problems + 4 essays
Double the question count of the supplied April 2026 FYUGP prerequisite paper.

CU-FYUGP syllabus 2024 · study edition
SI units · phone pages · October 2026

Practice instructions

Suggested time: two sittings of two hours. Section A: 20 questions × 3 marks, ceiling 48. Section B: 16 × 6, ceiling 72. Section C: answer any two of four × 10. Maximum: 140.

This doubles the supplied April 2026 **Electrodynamics I** 70-mark practice pattern. The current ED II syllabus specifies 70 theory marks but does not supply a question pattern here. This is a study paper, not an official FYUGP mock.

Each numbered question has a distinct primary assessment target. Longer questions combine earlier methods. N25 means November 2025 CBCSS ED I; A26 means April 2026 FYUGP ED I. Tags explicitly identify these older or prerequisite courses. Expanded/adapted means wording, scope or mark allocation was changed. Overlapping past questions are tagged with their original course and number; untagged questions are original model practice.

All four theory modules are included in the set. Module V is laboratory work and is outside this theory-paper series.

Section A · Short answers

3 marks each · ceiling 48

1. Define the LR time constant, state its unit and explain its growth interpretation.

Model · syllabus-based

2. Why does current in an ideal inductor remain continuous when a finite DC voltage is suddenly applied?

   Model · syllabus-based

3. During LR decay, how much magnetic energy remains when the current is half its initial value?

   Model · syllabus-based

4. Why must the LR decay formula specify a closed discharge path?

   Model · syllabus-based

5. An RC circuit has R=20 kΩ and C=50 μF. Find τ and the charging fraction at t=τ.

   Model · syllabus-based

6. Which capacitor quantity remains continuous under finite current at switching, and why?

   Model · syllabus-based

7. Write the natural angular frequency and frequency of an ideal LC circuit, and distinguish the units.

   Model · syllabus-based

8. An isolated LC circuit starts with a charged capacitor and zero current. Where is its energy after a quarter-cycle?

   Model · syllabus-based

9. An uncharged capacitor and ideal inductor are connected in series to a DC source V at t=0. What is the maximum capacitor voltage?

Model · syllabus-based

10. State the peak emf of a rotating N-turn coil and explain its dependence on angular speed.

Model · syllabus-based

11. At what coil orientations are flux and emf maximum in an ideal AC generator?

Model · syllabus-based

12. Describe voltage, current and average power for a pure resistor driven by sinusoidal voltage.

Model · syllabus-based

13. State the reactance and phase relation for an ideal inductor on sinusoidal AC.

Model · syllabus-based

14. State the reactance and phase relation for an ideal capacitor on sinusoidal AC.

Model · syllabus-based

15. What does multiplying a phasor by j do, and why is j used rather than i here?

Model · syllabus-based

16. Why can magnitudes of resistor and inductor voltages not be added directly in a series RL circuit?

Model · syllabus-based

17. What are ideal input current and impedance at parallel LC resonance under an AC voltage source?

Model · syllabus-based

18. State the resonance condition and input impedance of a series LCR circuit.

Model · syllabus-based

19. State whether a series LCR circuit is capacitive or inductive below and above its resonance frequency.

Model · syllabus-based

20. Define power factor and distinguish real power from apparent power.

Model · syllabus-based

Section B · Problems and explanations

6 marks each · ceiling 72

21. Derive current growth in a series LR circuit connected to constant V at t=0, with i(0)=0.

Model · syllabus-based

22. A 12 V step drives R=6 Ω and L=3 H from zero current. Find τ, i and inductor voltage at t=0.50 s.

Model · syllabus-based

23. Derive LR decay from initial current I_o and show the total resistor heat equals the initial magnetic energy.

Model · syllabus-based

24. An LR circuit has L=2 H and R=10 Ω. How long does current take to fall to 10% of its initial value?

Model · syllabus-based

25. Derive q(t), i(t) and capacitor voltage when an initially uncharged series RC circuit is connected to DC V.

Model · syllabus-based

26. A 100 μF capacitor initially at 10 V discharges through 20 kΩ. Using i=dq/dt for its initially positive plate, find q, v and i after 2 s.

Model · syllabus-based

27. For charging an initially uncharged capacitor from an ideal constant source V through R, find source work, final stored energy and resistor heat.

Model · syllabus-based

28. An ideal isolated circuit has L=0.10 H, C=10 μF, initial capacitor voltage 20 V and zero current. Find q(t), i(t), frequency and peak current.

Model · syllabus-based

29. Derive the ideal series LC response to a DC step V with zero initial charge and current. Explain the voltage overshoot.

Model · syllabus-based

30. A 100-turn coil of area 0.020 m² rotates at 50 revolutions/s in a 0.10 T uniform field. Find angular speed, emf peak and rms emf.

Model · syllabus-based

31. Derive the impedances of an ideal inductor and capacitor using j and the common time factor $e^{j\omega t}$.

Model · syllabus-based

32. At an angular frequency where $X_L = 40\ \Omega$, a series RL circuit has R=30 Ω and source 100 V rms. Find Z, current, phase and real power.

Model · syllabus-based

33. A series RC circuit has R=60 Ω, $X_C = 80\ \Omega$ and source 100 V rms. Find current, phase, component voltages and power.

Model · syllabus-based

34. An ideal parallel LC has L=0.10 H and C=10 µF, driven at resonance by 10 V rms. Find frequency, branch currents and source current.

Model · syllabus-based

35. A series circuit has R=10 Ω, L=0.10 H and C=10 µF, with 10 V rms at resonance. Find frequency, current and L/C rms voltages.

Model · syllabus-based

36. A sinusoidal load takes 4 A rms from 230 V rms at power factor 0.75 lagging. Find real power, apparent power and phase angle.

Model · syllabus-based

Section C · Essays

10 marks each · answer any two

37. Derive and compare LR growth/decay and RC charge/discharge, emphasizing initial conditions, time constants and limiting behaviour.

Model · syllabus-based

38. Develop free LC oscillation and compare it with an initially uncharged LC circuit subjected to a DC step. Explain energy in both cases.

Model · syllabus-based

39. Use complex impedance/admittance to explain and contrast series LCR resonance and ideal parallel LC resonance, including current and energy flow.

Model · syllabus-based

40. Derive average power for sinusoidal voltage and current with phase difference, then discuss pure R, L, C and series LCR at resonance.

Model · syllabus-based

Coverage audit

40 numbered questions; 40 primary targets. The list below makes the intended distinction explicit.

Q1 · **T1** LR time constant

Q2 · **T1** Current continuity at switching

Q3 · **T2** LR half-current energy

Q4 · **T2** Discharge path condition

Q5 · **T3** RC time constant

Q6 · **T3** Capacitor continuity

Q7 · **T4** LC natural frequency

Q8 · **T4** Quarter-cycle energy exchange

Q9 · **T5** Ideal LC step overshoot

Q10 · **T6** Generator peak dependence

Q11 · **T6** Flux-emf timing

Q12 · **T7** Pure resistor phase

Q13 · **T7** Pure inductor phase

Q14 · **T7** Pure capacitor phase

Q15 · **T8** Meaning of j

Q16 · **T8** Phasor voltage addition

Q17 · **T9** Parallel resonance input

Q18 · **T10** Series resonance condition

Q19 · **T10** Above and below resonance

Q20 · **T11** Power factor definition

Q21 · **T1** LR step solution

Q22 · **T1** LR transient numerical

Q23 · **T2** LR decay solution and heat

Q24 · **T2** Decay timing with logarithm

Q25 · **T3** RC charge solution

Q26 · **T3** RC discharge sign and numerical

Q27 · **T3** RC charging energy budget

Q28 · **T4** Free LC solution and numerical

Q29 · **T5** Driven ideal LC step derivation

Q30 · **T6** Generator numerical

Q31 · **T7** Deriving reactive impedances

Q32 · **T8** Series RL phasor calculation

Q33 · **T8** Series RC phasor calculation

Q34 · **T9** Parallel LC branch currents

Q35 · **T10** Series LCR resonance and voltage magnification

Q36 · **T11** AC power numerical

Q37 · **T3** Comparing first-order transients

Q38 · **T4** Free versus driven LC and conservation

Q39 · **T10** Series and parallel resonance comparison

Q40 · **T11** Derivation of average AC power

Past-paper accounting

No question from the supplied local past papers maps directly to this module. All 40 are model questions, guided by the syllabus; no invented PYQ provenance.